

HYDROGENATED NBR: PATENT LANDSCAPE FOR SYNTHESIS, HYDROGENATION CONTROL, AND HNBR-CONTAINING ELASTOMER SYSTEMS

1. ABSTRACT

The provided patent set spans both direct hydrogenated NBR synthesis/manufacture and adjacent enabling technologies that use HNBR as a functional elastomer component. The core target is hydrogenated nitrile-butadiene rubber (HNBR), defined here as NBR-derived material with reduced residual unsaturation and improved thermal/oxidative stability, while retaining elastomeric performance. Among the manifest patents, EP2519350B1 is the most directly process-oriented: it discloses novel ruthenium metathesis catalysts and explicitly states their use in depolymerization of NBR/SBR and in hydrogenation of rubbers containing carbon-carbon double bonds, including making depolymerized HNBR or HNBR by sequential depolymerization/hydrogenation under high pressure hydrogen at 60–150°C. US20180022951A1 is formulation-oriented but technically relevant because it uses NBR and HNBR as the nitrile-rubber phase in stretchable electroconductive silver pastes; it emphasizes low alkali metal content, emulsion-polymerized rubber, and HNBR with high nitrile content. EP4221260B1 is a downstream elastomer-composite disclosure in which nitrile rubber and hydrogenated nitrile rubber are blended in a diaphragm rubber compound, with acrylonitrile block content controlled at 20–80 wt%, filler at 5–60 wt%, and vulcanizing agent at 0.5–5 wt% to balance damping, resilience, and mechanical strength. The remaining manifest patents are not HNBR synthesis patents: US10927052B2 concerns branched decene oligomers and subsequent hydrogenation of olefin oligomers; US8247063B2 concerns multilayer materials and mentions hydrogenated rubbers only as generic examples; EP3168256B1 concerns polyurethane eyewear materials based on cyclohexane diisocyanate chemistry; EP2273890B1 is unrelated confectionery. Overall, the evidence indicates that HNBR-related patent activity in this set is dominated by catalyst-enabled hydrogenation/depolymerization chemistry and by formulation strategies that exploit HNBR's polarity, thermal stability, and low-temperature resilience rather than by detailed standalone NBR polymerization routes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2273890B1
- Patent Title: Sugar free mannitol confectionery and methods of making same
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example example 4: - Example Extracted Example example 4: - Example Extracted Example Example 1:: - Example Extracted Example Example 2:: - Example Extracted Example Example 3:: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 4:: - Example Extracted Example Example 2.: - Example Extracted Example Example 5.: - Example Extracted Example Example 4: - Example Extracted Example Example 4.: - Example Extracted Example Example 2.: - Example Extracted Example Example 4: - Example Extracted Example Example 6.: - Example Extracted Example Example 4: - Example Extracted Example Example 4.: - Example Extracted Example Example 2.:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 3: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 4: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 5: - Example Extracted Example Example 2: - Example Extracted Example Example 6: - Example Extracted Example

Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 7: -
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Extracted Example Example 8: - Example Extracted Example Example 1: - Example Extracted
Example Example 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US10927052B2
- Patent Title: Decene oligomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted
Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example
Example 4: - Example Extracted Example Example 3: - Example Extracted Example Example 4.: -
Example Extracted Example Example 5: - Example Extracted Example Example 5: - Example
Extracted Example Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US8247063B2
- Patent Title: Multilayer material and related methods
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 2:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US20180022951A1
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 2: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 3: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 4: - Example Extracted Example Preparation Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: EP3168256B1
- Patent Title: Eyewear material comprising a polyurethane resin, eyewear frame and lens
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example

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Extracted Example Example 2:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP4221260B1
- Patent Title: MEMBRAN AND SOUND GENERATION DEVICE
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2519350B1 is the only primary patent in the manifest that explicitly discloses a hydrogenation route to HNBR; it describes depolymerization of NBR/SBR at 30–100°C and hydrogenation under high-pressure hydrogen at 60–150°C, with catalyst addition order varying between depolymerization-first and hydrogen-first sequences.
- EP2519350B1 focuses on transition-metal metathesis catalysts (Ru complexes) rather than NBR feed composition; no monomer feed ratios for NBR polymerization are disclosed in the retained evidence, because the patent is about post-polymerization modification/depolymerization rather than NBR synthesis.
- US20180022951A1 uses NBR/HNBR as a rubber phase in conductive paste formulations; the retained examples show NBR with nitrile group contents of 26–51 wt% and HNBR with nitrile group content of 50 wt% and iodine value 25 mg/100 mg, but it does not disclose HNBR

synthesis conditions.

- US20180022951A1 emphasizes low alkali metal content in the nitrile rubber phase, with the claim threshold at 4,000 ppm or less and preferred levels at 1,500 ppm or less; the examples show performance degradation at 5,500 ppm, supporting the importance of purification/washing after emulsion polymerization.
- EP4221260B1 is formulation-focused and does not disclose HNBR synthesis; it specifies a rubber compound containing both nitrile rubber and hydrogenated nitrile rubber, with acrylonitrile block content of 20–80 wt% and a nitrile rubber/HNBR mass ratio of at least 1.
- EP4221260B1 differs from US20180022951A1 in end use and property targets: it prioritizes damping factor, tensile strength, tear strength, and temperature resistance for diaphragms, whereas US20180022951A1 prioritizes low resistance, extensibility, repetitive stretchability, and anti-migration in conductive films.
- Across the primary patents, the recurring theme is that HNBR is valued for high polarity and thermal stability, but the evidence splits into two approaches: catalyst-enabled chemical transformation of NBR (EP2519350B1) and property-driven blending/formulation of HNBR with other rubbers or fillers (US20180022951A1, EP4221260B1).
- No primary patent in the manifest provides a detailed NBR monomer feed ratio for direct NBR polymerization as the central invention; the only explicit NBR composition data in the retained examples are post-polymerization rubber characteristics such as nitrile content and alkali metal content.
- Historical versus newer approach: the older, more conventional route reflected in US20180022951A1 relies on emulsion-polymerized NBR followed by hydrogenation and purification, while EP2519350B1 proposes novel Ru catalysts for more selective depolymerization/hydrogenation operations, and EP4221260B1 leverages HNBR as a component in engineered rubber compounds rather than as a synthesis target.

4. CONCLUSION

The evidence supports a two-part technical picture for hydrogenated NBR. First, the most direct synthesis/manufacturing disclosure in the manifest is EP2519350B1, which teaches Ru-catalyzed depolymerization of NBR followed by hydrogenation under high-pressure hydrogen at 60–150°C, or hydrogen-first followed by catalyst addition, as a route to depolymerized HNBR/HNBR. Second, the other HNBR-relevant patents are formulation patents that exploit HNBR's properties rather than teach its preparation: US20180022951A1 shows that low-alkali, emulsion-polymerized NBR/HNBR improves conductive paste stability and stretchability, while EP4221260B1 shows that blending nitrile rubber with HNBR and controlling acrylonitrile block content, filler loading, and vulcanization can deliver damping and mechanical balance. Based on the evidence, the most defensible synthesis guidance is to start from a purified emulsion-polymerized NBR precursor, minimize residual alkali metal by washing, and perform selective hydrogenation under high-pressure hydrogen with a catalyst system capable of operating in the 60–150°C window. For downstream elastomer design, the evidence favors maintaining substantial acrylonitrile content, using HNBR as a co-phase with NBR when cost or damping balance is needed, and controlling

filler and cure package to tune resilience and thermal performance.

5. REFERENCES

- EP2273890B1 | Sugar free mannitol confectionery and methods of making same | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2273890B1/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US10927052B2 | Decene oligomers | Not disclosed | US | 2021 | <https://patents.google.com/patent/US10927052B2/en>
- US8247063B2 | Multilayer material and related methods | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8247063B2/en>
- US20180022951A1 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180022951A1/en>
- EP3168256B1 | Eyewear material comprising a polyurethane resin, eyewear frame and lens | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3168256B1/en>
- EP4221260B1 | MEMBRAN AND SOUND GENERATION DEVICE | Not disclosed | EP | 2025 | <https://patents.google.com/patent/EP4221260B1/en>